

Productivity, Output, and Employment*

Student Notes

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Learning objectives. Read the production function and its slope (MPK, MPN); derive labour demand from $w = MPN$; split a wage change into income and substitution effects; clear the classical labour market to get $\tilde{N}$ and $\tilde{Y}$; measure unemployment and its flows; read Okun's law.

3.1 The Production Function

$$Y = A F(K, N) \quad (3.1)$$

Inputs: capital K , labour N . **A :** total factor productivity (TFP), the effectiveness with which K and N are used. A is never observed directly; it is backed out as a residual, the **Solow residual**. (Solow, 1957; in 1987, the year of his Nobel, he quipped that “you can see the computer age everywhere but in the productivity statistics.”)

Two properties: the curve slopes **up** (more input, more output) and is **concave** (its slope falls as the input grows). That falling slope is the *diminishing marginal product*.

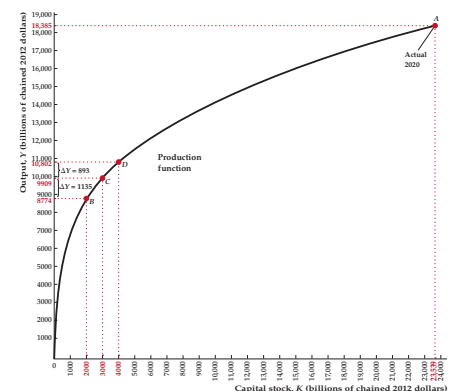


Figure 3.1: Two properties. Upward sloping (more input, more output) and concave: the slope, = marginal product, falls as the input rises. Hold A and K fixed to draw Y against N .

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3.1.1 The Cobb–Douglas form

$$Y = AK^\alpha N^\beta \quad (3.2)$$

The workhorse specification. α, β are the **output elasticities** of capital and labour: a 1% rise in K lifts Y by $\alpha\%$; a 1% rise in N lifts Y by $\beta\%$. (Douglas noticed labour's income share sitting near two-thirds and in 1927 asked the mathematician Charles Cobb for a formula to fit it.)

Property	What it says
Returns to scale	Degree of homogeneity = sum of elasticities: scale all inputs by λ and $Y \rightarrow \lambda^{\alpha+\beta}Y$. So $\alpha + \beta \gtrless 1$ gives increasing / constant / decreasing returns.
Marginal products	$MPK = \alpha AK^{\alpha-1}N^\beta$, $MPN = \beta AK^\alpha N^{\beta-1}$; both positive and diminishing when $0 < \alpha, \beta < 1$.
Income-share duality	Under competition (factors paid their marginal product) labour's share $wN/Y = \beta$ and capital's share $rK/Y = \alpha$: <i>elasticities equal shares</i> .
Elasticity of substitution	Fixed at the knife-edge value $\sigma = 1$; factor shares stay put as relative supplies move.
Essential factors	No factor fully substitutes for another, so the domain is the strictly positive reals ($K, N > 0$).

Table 3.1: Cobb–Douglas: the minimal properties economists want, at a price. (Palgrave: Cobb–Douglas)

Restrictive (that $\sigma = 1$ is a knife edge) and sometimes dismissed as a toy, yet the Cobb–Douglas is a special case of richer forms: take the CES $[\sum a_i x_i^t]^{1/t}$ and let $t \rightarrow 0$ and it collapses to Cobb–Douglas; it also nests inside the translog. That robustness is why it endures. (Palgrave: Cobb–Douglas)

Example 3.1.1 (Constant returns and the two-thirds share) Take $Y = AK^{0.33}N^{0.67}$, a good fit for the EU economy. Then $\alpha + \beta = 1$: doubling K and N exactly doubles Y ,

$$A(\lambda K)^{0.33}(\lambda N)^{0.67} = \lambda^{0.33+0.67} AK^{0.33}N^{0.67} = \lambda Y.$$

Labour's income share is 0.67 and capital's is 0.33, matching the elasticities exactly. This steady labour share has a name, **Bowley's Law**; Keynes called its stability "one of the most surprising, yet best-established, facts in the whole range of economic statistics" (it has slipped somewhat in recent decades). (class slides)

3.1.2 Returns to scale

Scale *every* input by the same factor $m > 1$ ($m = 2$ doubles both K and N) and compare the new output $Y' = AF(mK, mN)$ with $m \times Y$: a rise of *more* than m is **increasing** returns, of *exactly* m **constant**, of *less* **decreasing**.

$$Y = AK^\alpha N^\beta \Rightarrow Y' = A(mK)^\alpha (mN)^\beta = m^{\alpha+\beta} Y$$

$$\alpha + \beta > 1 \text{ increasing} \quad \alpha + \beta = 1 \text{ constant} \quad \alpha + \beta < 1 \text{ decreasing}$$

The exponent sum *is* the power output is raised to, so a single number settles it; and because Cobb–Douglas is homogeneous of degree $\alpha + \beta$ the verdict is **global** (the same at every output level). Put $m = 2$ to size it up:

Two traps. (i) **Long vs short run:** returns to scale moves *all* inputs together; the diminishing marginal product above holds the other input *fixed*. (ii) **Independent:** $\alpha = 0.6, \beta = 0.7$ has increasing returns ($\alpha + \beta = 1.3$) yet diminishing MPK, MPN (each exponent < 1).

Why the exponent adds. Put $K \rightarrow mK$, $N \rightarrow mN$ in $Y = AK^\alpha N^\beta$:

$$\begin{aligned} Y' &= A(mK)^\alpha (mN)^\beta \\ &= m^\alpha m^\beta AK^\alpha N^\beta \\ &= m^{\alpha+\beta} Y, \end{aligned}$$

using $(mK)^\alpha = m^\alpha K^\alpha$ and $m^\alpha m^\beta = m^{\alpha+\beta}$. Inputs scale by $m = m^1$, output by $m^{\alpha+\beta}$: compare $\alpha + \beta$ with 1.

output scales by $2^{\alpha+\beta}$, so $\alpha+\beta = 1$ gives $2^1 = 2$ (output doubles), $\alpha+\beta = 1.2$ gives $2^{1.2} \approx 2.30$ (a 130% rise — *more than double*), and $\alpha+\beta = 0.8$ gives $2^{0.8} \approx 1.74$ (only 74%). **When $\alpha+\beta > 1$** the advantage compounds — each further doubling of both inputs multiplies output by more than two, so a firm that keeps scaling pulls steadily ahead (the technical root of a tendency to monopoly). The mirror holds **when $\alpha+\beta < 1$** : the output multiplier $m^{\alpha+\beta}$ now *trails* the input multiplier m ($2^{0.8} < 2$), so doubling inputs less-than-doubles output and each further doubling slips a little further behind. The single logic behind all three is a **race of exponents** — the inputs carry the exponent 1 (you scaled them by $m = m^1$) while output carries $\alpha+\beta$, so output outpaces, matches, or trails the inputs exactly as $\alpha+\beta$ sits above, at, or below 1. Constant returns is the natural benchmark: you can always *replicate* the whole plant and get double the output, so increasing returns need something unreplicable (specialisation, indivisibilities) and decreasing returns usually mean a fixed factor was left un-scaled. Under constant returns, paying each factor its marginal product just exhausts output (Euler's theorem), which is why the elasticities equal income shares above. (class slides)

3.1.3 Slopes: MPK and MPN

The marginal product of an input is the **slope** of the production function against that input, holding the other input and A fixed. Both MPK and MPN are positive and both diminish (the curve flattens as the input grows). MPN is the one that will become labour demand.

3.1.4 Supply shocks

A **supply (productivity) shock** shifts the whole production function. Positive (invention, good weather): A up, curve up, slope up. Adverse (bad weather, an oil-price spike, new regulation): A down, curve down, slope down. The flatter slope, a lower MPN , is what will shift labour demand. ("Stagflation" is exactly an adverse supply shock: output down, prices up.)

3.2 The Demand for Labor

Setup: capital fixed (short run), workers alike, competitive market, firms maximise profit. Hire one more worker while the benefit beats the cost.

Real rule:	$w = MPN$
Nominal rule:	$W = MRPN = P \times MPN \quad (3.3)$

Same rule, two units: divide the nominal rule by P and $W/P = MPN$, i.e. $w = MPN$. Keep units matched, money vs money (W vs $MRPN$) or real vs real (w vs MPN). If $w > MPN$ the last worker is overpaid, cut N ; if $w < MPN$, add N ; profit is highest at $w = MPN$.

Example 3.2.1 (Clip Joint / RKn Widget: reading labour demand off MPN) Output rises with workers but by less each time (diminishing MPN). With price $P = \$5$ a widget, the value of the marginal worker is $MRPN = P \times MPN$. Hire while $MRPN \geq W$.

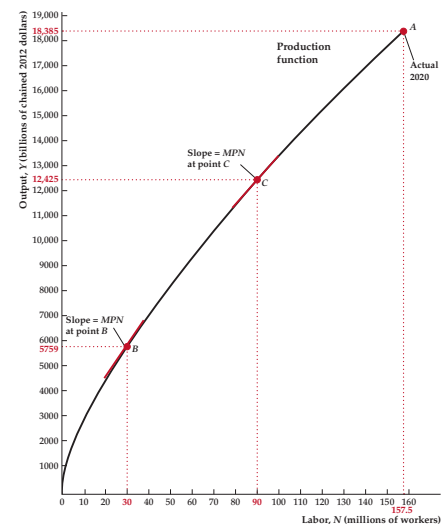


Figure 3.3: $MPN = \Delta Y / \Delta N$ is the slope of Y against N (and $MPK = \Delta Y / \Delta K$ the slope against K); both fall as the input rises

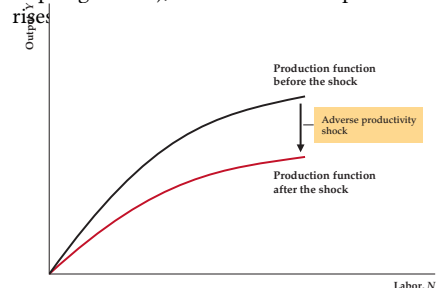


Figure 3.4: An adverse shock lowers output *and* the slope at every N : MPN falls.

Workers N	Output	MPN	$MRPN = 5 MPN$
0	0	—	—
1	8	8	\$40
2	15	7	\$35
3	21	6	\$30
4	26	5	\$25
5	30	4	\$20
6	33	3	\$15

At $W = \$38$ hire **1** (only worker 1 clears \$38); at $W = \$27$ hire **3**; at $W = \$22$ hire **4**. Lower the wage, hire more: the $MRPN$ schedule is the (nominal) labour demand curve.

The labour demand curve is the MPN curve, downward sloping. A **wage** change is a move *along* it. What **shifts** it:

An increase in	Shifts ND	Because
Productivity (A)	right	beneficial shock raises MPN
Capital stock (K)	right	more K raises MPN

Table 3.2: What shifts labour demand. (A wage change does not: it moves along the curve.)

3.3 The Supply of Labor

Households choose between **consumption** (needs income, needs work) and **leisure**. Work another day while its benefit (more consumption) exceeds its cost (lost leisure); stop when they are equal.

Effect of a wage rise	Direction
Substitution effect	Leisure is dearer (each hour off costs more wages) $\Rightarrow$ work <i>more</i> .
Income effect	Same hours now buy more $\Rightarrow$ afford more leisure $\Rightarrow$ work <i>less</i> .
Temporary rise (pure substitution)	Wealth barely moves $\Rightarrow$ work more; labour supply rises.
Permanent rise	Income effect bites $\Rightarrow$ labour supply can fall.
Lottery win (pure income)	No change in the reward to work $\Rightarrow$ work less.

Table 3.3: One wage rise, two hands: substitution pulls toward work, income away from it. (Real lottery winners cut work by about a dime per dollar won: the pure income effect.)

Shifts NS left (less supply): higher wealth; higher expected future wage.
Shifts NS right (more supply): lower wealth; lower expected future wage; a larger working-age population (births, immigration); higher participation. Aggregate NS slopes up through two margins, the *intensive* (hours per worker) and the *extensive* (whether to work at all); the extensive margin makes it far more elastic than micro hours-studies suggest. (Keane & Rogerson 2012)

3.4 Labor Market Equilibrium

Classical model: the real wage adjusts fast, so the market clears. The crossing pins down the **full-employment** employment $\bar{N}$ and the market-clearing wage w^* . Its limitation: everyone who wants to work at w^* does, so there is no involuntary unemployment.

$$\bar{Y} = A F(K, \bar{N}) \quad (3.4)$$

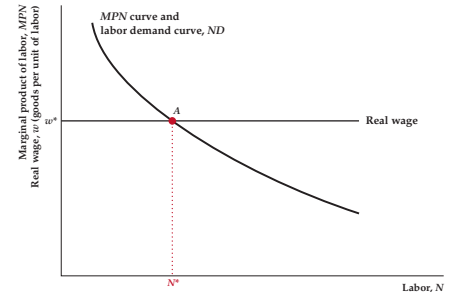


Figure 3.5: Marginal analysis: hire the worker if $MRPN \geq W$ (equivalently $MPN \geq w$). Profit is highest where they are equal.

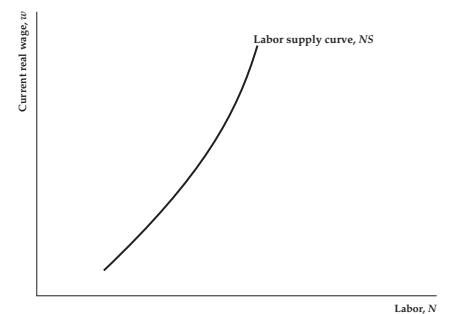


Figure 3.7: Individual labour supply: upward over the normal range; it bends back only at very high wages.

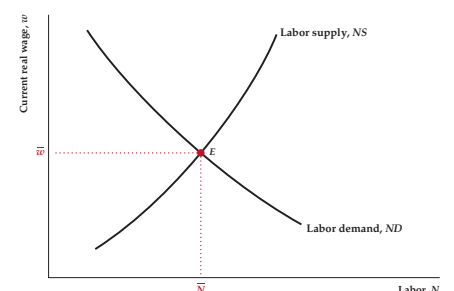


Figure 3.9: Equilibrium E : $ND = NS$ fixes the full-employment $\bar{N}$ and the market-clearing wage w^* .

Full-employment (potential) output $\bar{Y}$: plug $\bar{N}$ back into the production function. It moves when $\bar{N}$ moves or the production function shifts (a supply shock).

A temporary adverse supply shock lowers MPN , shifts ND left, and slides the equilibrium down: the real wage falls and $\bar{N}$ falls, so $\bar{Y}$ falls. The 1973–74, 1979–80, and 2003–08 oil-price spikes are the textbook cases; a rule of thumb is that a 10% oil-price rise cuts GDP by about 0.4 percentage points. (class slides)

3.5 Unemployment

The classical market clears, so it cannot house unemployment. To measure it, split the working-age population three ways.

Labour force	= Employed + Unemployed
Unemployment rate	= Unemployed / Labour force
Participation rate	= Labour force / Working-age pop.
Employment ratio	= Employed / Working-age pop.

Unemployed = no job, available, and *actively searched* in the last four weeks. **Discouraged workers** stop searching, so they leave the labour force and the headline rate; that is why a higher jobless rate can come with a *lower* participation rate. The pain is more than lost income: unemployment lowers mental and physical health even after controlling for income.

Type	Source
Frictional	Search and matching take time (heterogeneous jobs and workers).
Structural	Long-term / chronic mismatch, present even outside recessions.
Natural rate $\bar{u}$	= frictional + structural, when output is at $\bar{Y}$.
Cyclical	$u - \bar{u}$: the gap over the business cycle.

Table 3.4: Types of unemployment. Most *spells* are short, yet most people unemployed on a given day are in long spells. (The natural rate is Friedman's, 1968: no permanent unemployment–inflation trade-off to exploit.)

Six official measures, U-1 to U-6 (the headline is U-3):

- U-1 unemployed ≥ 15 weeks
- U-2 job losers + completed temporary jobs
- U-3 **official rate**
- U-4 U-3 + discouraged workers
- U-5 U-4 + marginally attached workers
- U-6 U-5 + part-time for economic reasons

3.6 Okun's Law

$$\frac{\bar{Y} - Y}{\bar{Y}} = 2(u - \bar{u}) \quad (3.5)$$

$$\frac{\Delta Y}{Y} = 3 - 2\Delta u \quad (3.6)$$

Each 1 percentage point of cyclical unemployment costs about 2% of output, not 1%: when unemployment rises the labour force also shrinks, hours per worker fall, and average productivity dips. With no change in u , output grows at its $\approx 3\%$ trend. It is a reliable regularity, not physics (2009 bent it).

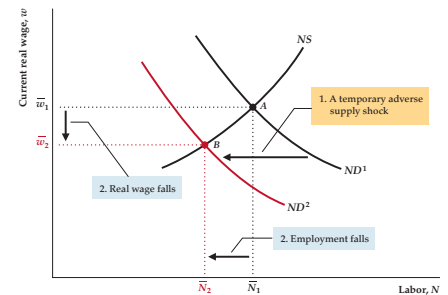


Figure 3.10: Adverse supply shock. ND shifts left; w^* and $\bar{N}$ both fall, so $\bar{Y}$ falls.

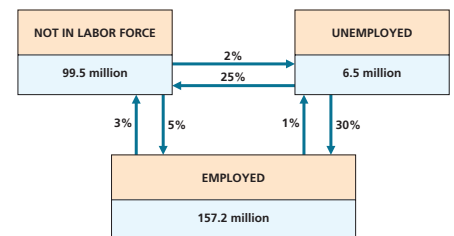


Figure 3.12: Flows in a typical month between employed, unemployed, and not-in-the-labor-force. Stocks change only when the flows are unequal.

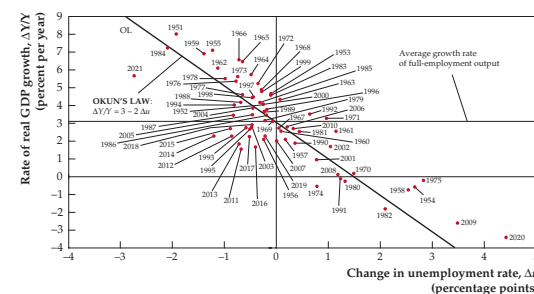


Figure 3.16: Okun's law, US: output growth against the change in unemployment. The line has slope -2 ; the 3% line is trend growth.

3.7 Quick reference (cheat sheet)

3.7.1 Symbols

$Y, \bar{Y}$	output; full-employment (potential) output
$K, N, \bar{N}$	capital; labour; full-employment labour
A	total factor productivity (Solow residual)
α, β	output elasticities = income shares (Cobb–Douglas)
MPK, MPN	marginal product of capital, of labour
w, W	real wage = W/P ; nominal wage
$MRPN$	marginal revenue product = $P \times MPN$
ND, NS	labour demand (= MPN), labour supply
$u, \bar{u}$	unemployment rate; natural rate

3.7.2 Distinctions that trap people

- **Move vs shift.** A wage change is a move *along* ND/NS ; a shock, capital, or wealth shifts the curve.
- **MPN vs $MRPN$.** MPN is extra output (units); $MRPN = P \times MPN$ is extra revenue (money). Compare like with like.
- **Level vs slope.** A supply shock changes both the *height* of the production function and its *slope* (MPN); the slope change is what moves labour demand.
- **Voluntary vs involuntary.** The classical market clears, so its jobless are voluntary; frictional/structural/cyclical unemployment live outside it.

3.7.3 The chain

Slope of the production function is $MPN \rightarrow MPN$ is labour demand $\rightarrow$ demand meets supply at $\bar{N} \rightarrow \bar{N}$ gives $\bar{Y} \rightarrow$ deviations of Y from $\bar{Y}$ are Okun's law.

3.7.4 Key relations

$$Y = AF(K, N) \quad (3.1)$$

$$Y = AK^\alpha N^\beta \quad (3.2)$$

$$MPK = \alpha AK^{\alpha-1} N^\beta$$

$$MPN = \beta AK^\alpha N^{\beta-1}$$

$$W = MRPN = P \times MPN \quad (3.3)$$

$$w = MPN \text{ at the optimum}$$

$$\bar{Y} = AF(K, \bar{N}) \quad (3.4)$$

$$(\bar{Y} - Y)/\bar{Y} = 2(u - \bar{u}) \quad (3.5)$$

$$\Delta Y/Y = 3 - 2\Delta u \quad (3.6)$$

Sources: Figures reproduced from Abel, Bernanke & Croushore, *Macroeconomics*, 11th ed. Acemoglu, D., Laibson, D., & List, J., *Macroeconomics* (Global ed.), Pearson. Mankiw, N. G. (2023), *Macroeconomics*, 12th ed., Worth. “Cobb–Douglas Functions,” *The New Palgrave Dictionary of Economics*. Keane, M., & Rogerson, R. (2012). The main text follows Abel, Bernanke & Croushore and is not cited in the body.